

# IRIS FPU data sheet and design overview

## 1 Overview

The IRIS FPU is an IEEE-754 floating point unit implementing all floating point operations as required by the SPARC V8 standard. It supports both single- and double-precision operations including de-normalized numbers. All four IEEE-754 rounding modes are supported. The interface to the FPU has been modeled after the Meiko FPU from the Sun Microsparc processor. The IRIS FPU is iterative, only one instruction is calculated at a time.

## 2 Instructions

The IRIS FPU implements all necessary FPU instructions required by the SPARC V8 standard.

| Instruction | Opcode | Op1 | Op2   | Result | Exceptions         | Description                               |
|-------------|--------|-----|-------|--------|--------------------|-------------------------------------------|
| FADDS       | 041    | SP  | SP    | SP     | NV, OF, UF, NX     | Add single                                |
| FADDD       | 042    | DP  | DP    | DP     | NV, OF, UF, NX     | Add double                                |
| FSUBS       | 045    | SP  | SP    | SP     | NV, OF, UF, NX     | Subtract single                           |
| FSUBD       | 046    | DP  | DP    | DP     | NV, OF, UF, NX     | Subtract double                           |
| FMULS       | 049    | SP  | SP    | SP     | NV, OF, UF, NX     | Multiply single                           |
| FMULD       | 04A    | DP  | DP    | DP     | NV, OF, UF, NX     | Multiply double                           |
| FSMULD      | 069    | SP  | SP    | DP     | NV                 | Multiply single to double                 |
| FDIVS       | 04D    | SP  | SP    | SP     | NV, OF, DZ, UF, NX | Divide single                             |
| FDIVD       | 04E    | DP  | DP    | DP     | NV, OF, DZ, UF, NX | Divide double                             |
| FSQRTS      | 029    | -   | SP    | SP     | NV, NX             | Square root single                        |
| FSQRD       | 02A    | -   | DP    | DP     | NV, NX             | Square root double                        |
| FITOS       | 0C4    | -   | INT32 | SP     | NX                 | Convert 32-bit integer to single          |
| FITOD       | 0C8    | -   | INT32 | DP     | -                  | Convert 32-bit integer to double          |
| FSTOI       | 0D1    | -   | SP    | INT32  | NV, NX             | Convert single to 32-bit integer          |
| FDTOI       | 0D2    | -   | DP    | INT32  | NV, NX             | Convert double to 32-bit integer          |
| FSTOD       | 0C9    | -   | SP    | DP     | NV                 | Convert single to double                  |
| FDTOS       | 0C6    | -   |       |        | NV, OF, UF, NX     | Convert double to single                  |
| FCMPS       | 251    | SP  | SP    | CC     | NV                 | Compare single                            |
| FCMPD       | 252    | DP  | DP    | CC     | NV                 | Compare double                            |
| FCMPES      | 255    | SP  | SP    | CC     | NV                 | Compare single and exception if unordered |
| FCMPED      | 256    | DP  | DP    | CC     | NV                 | Compare double and exception if unordered |
| FABSS       | 009    | -   | SP    | SP     | -                  | Absolute value single                     |
| FNEGS       | 005    | -   | SP    | SP     | -                  | Negate single                             |
| FMOVS       | 001    | -   | SP    | SP     | -                  | Move (copy) single                        |

### 3 Signals

The table below list the IRIS FPU signals. All signals are active high. All inputs are sampled on the rising edge of CLOCK, while all outputs are driven on the rising edge of CLOCK.

| Signal              | I/O | Description                                                                                                                               |
|---------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------|
| CLOCK               | I   | Clock                                                                                                                                     |
| FPINST[9:0]         | I   | FPU instruction opcode                                                                                                                    |
| FPOP                | I   | Load FPU opcode                                                                                                                           |
| FPLD                | I   | Load FPU operands and start execution                                                                                                     |
| RESET               | I   | Synchronous reset, active high                                                                                                            |
| FPRF_DOUT1[63:0]    | I   | Operand 1                                                                                                                                 |
| FPRF_DOUT2[63:0]    | I   | Operand 2                                                                                                                                 |
| ROUNDINGMODE        | I   | Rounding mode.<br>00 – round to nearest, 01 – round to zero, 10 – round to +inf, 11 – round to -inf                                       |
| FPBUSY              | O   | Indicates the the FPU is executing                                                                                                        |
| FRACRESULT[54:3]    | O   | Fraction result                                                                                                                           |
| EXPRESULT[10:0]     | O   | Exponent result                                                                                                                           |
| SIGNRESULT          | O   | Sign result                                                                                                                               |
| SNNOTDB             | O   | Indicates that results is single precision or 32-bit integer                                                                              |
| EXCEP[5:0]          | O   | FPU exceptions                                                                                                                            |
| CONDITIONCODES[1:0] | O   | Result (condition codes) of a compare instruction<br>00 – equal<br>01 – operand1 < operand2<br>10 – operand1 > operand2<br>11 - unordered |

Single-precision operands should be placed in FPRF\_DOUT1/2[63:32]. If the result is single-precision or 32-bit integer (SNNOTDB = 1), the value is given by:

Result = SIGNRESULT & EXPRESULT[7:0] & FRACRESULT[54:32].

If the result is double-precision (SNNOTDB=0), the value is given by:

Result = SIGNRESULT & EXPRESULT[10:0] & FRACRESULT[54:3].

If an instruction only takes one operand, it should be placed on FPRF\_DOUT2.

## 4 Generics

The operation of the IRIS FPU can be configured through five generics.

| Generic     | Default value | Description                                                                                                                                                                                                                                                                                                                                                           |
|-------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fpga        | 0             | If set to 1, the sticky bit generation is implemented with an adder. This can improve timing on FPGA technologies but should not be used on ASICs.                                                                                                                                                                                                                    |
| Busydelay   | 0             | The delay in clock cycles from FpBusy to valid result data. Best timing is achieved with setting this generic to 0, which indicates that FpBusy is driven low in the same cycle as the result data is valid. If set to 1, FpBusy is driven low one clock before the result data is valid. Should only be set to 1 if direct compatibility with the Meiko is required. |
| Srtopt      | 3             | Indicates how many bits the SRT divider can evaluate per cycle. A value of 3 provides highest throughput, while lower numbers (0 – 2) will reduce area and throughput of the divider.                                                                                                                                                                                 |
| TinyPostRnd | 0             | If set to 1, the FPU will detect tininess after rounding otherwise before. For Meiko compatibility, set to 0, otherwise IEEE-754 recommends 1 (post rounding).                                                                                                                                                                                                        |
| FsMuld      | 0             | If set to 1, will enable the FSMULD instruction. If set to 0, an FSMULD instruction will cause an unimplemented trap (%fsr.ftt = 3).                                                                                                                                                                                                                                  |

## 5 Instruction latency

The instruction latency is measured from the assertion of FpOp to FpBusy low. The latency is somewhat data dependent with the minimum, maximum and average timing shown in the table below.

| Instruction               | Description              | Latency min. | Latency avg. | Latency max. |
|---------------------------|--------------------------|--------------|--------------|--------------|
| FADDS, FADD, FSUBS, FSUBD | Add/Sub single/double    | 5            | 5.5          | 9            |
| FMULS, FSMULD             | Multiply single          | 5            | 7.5          | 11           |
| FMULD                     | Multiply double          | 5            | 8.5          | 12           |
| FDIVS                     | Divide single            | 11           | 21           | 32           |
| FDIVD                     | Divide double            | 26           | 31           | 63           |
| FSQRTS                    | Square root single       | 34           | 34           | 35           |
| FSQRD                     | Square root double       | 61           | 61           | 62           |
| FSTOD                     | Convert single to double |              | 5            |              |
| FDTOS                     | Convert double to single |              | 5            | 8            |

## 6 Design notes

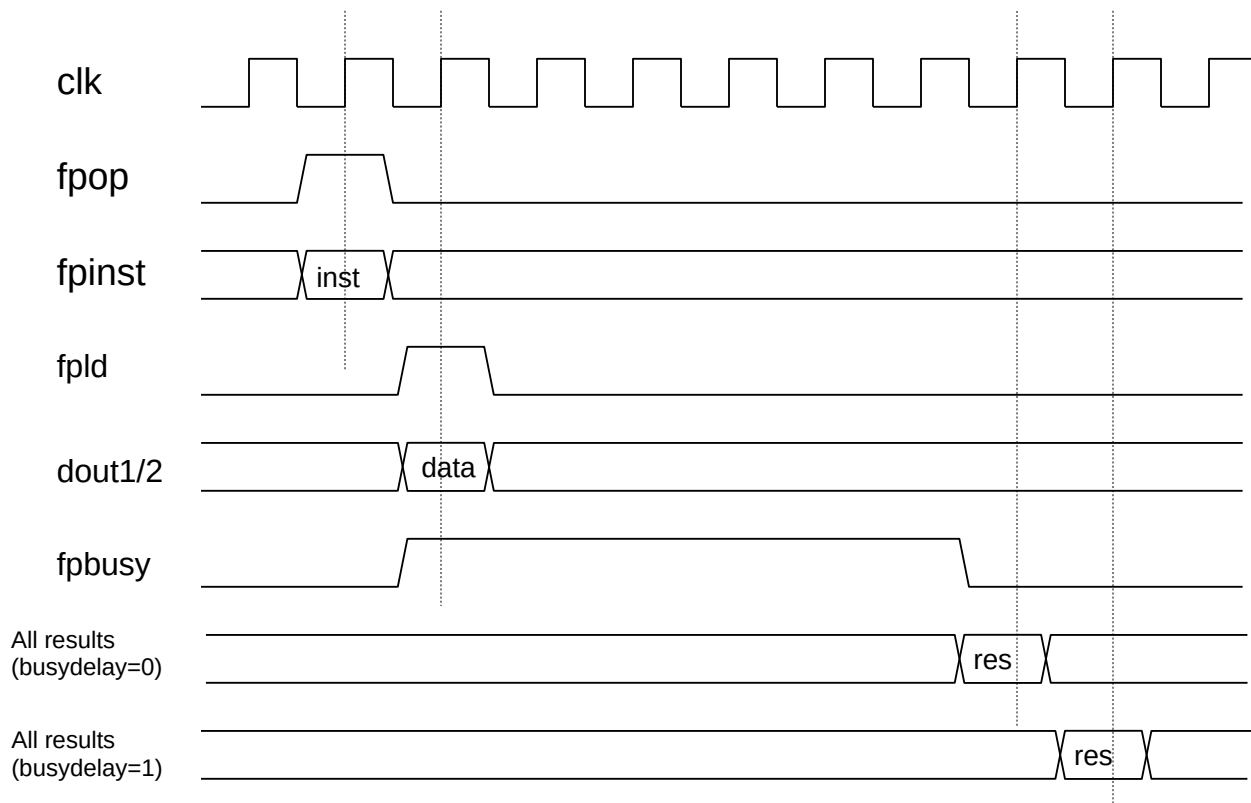
### 6.1 Generation of underflow exception (*unf*)

Underflow (tininess) can be detected before or after final rounding by setting the TinyPostRnd generic. This should be taken into account when generating test vectors with software like TestFloat.

### 6.2 Convert Floating point to Integer Instructions (FDTOI, FSTOI)

These instructions convert the floating-point operand on input 2 into a signed 32-bit integer. The result is always rounded toward zero (the RM input is ignored). On positive overflow, MAX INT is returned (0x7FFFFFFF) while on negative overflow, MIN INT (0x80000000) is returned.

## 7 Timing diagram



## 8 Area and timing

The IRIS FPU occupies approximately 3,500 LUT and 3 DSP blocks (18x18 multipler) on Xilinx FPGAs. The maximum frequency ranges from 100 MHz on Spartan6 to 200 MHz on Virtex7.